

Amine's Theory of Powers & Dimensions

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Preface

“And just as a disclaimer, the try is completely innovative and bold, so please before you start judging please change your lenses ;). Thanks in advance my dears!

Also note: I AM NOT A PROFESSIONAL MATHEMATICIAN AND THAT THIS IS ONLY AMATEUR TEENAGE WORK DONE WITH VERY LITTLE RESSOURCES, anyway sending LOVE!”

Abstract

This is in some kind a simple and innocent try, from a random kid, to think outside of the box and come out with new ideas, that may be right or wrong, who knows? Only GOD does. And let's not forget honourable mentions, thanks for my bus buddy again on this one, because he might have actually helped refine this paper especially, my old Nokia – RIP –, and let's especially not forget my pens and papers: Sending LOVE. Here we go again..

Introduction

In this modest paper, I will introduce my also modest theory that modestly tries to link math and physics through relating powers (math) and dimensions (physics), and the hot part: to revive Euler's Sum of Powers back from the coffin! Before bombing my house, here me out...

I know that Euler's Sum of Powers states that in order to express the following sum

$$\sum_{i=0}^n x_i^n$$

you need at least n terms which has been proven for $n = 2$ and is believed to also be true for $n = 3$.

And I also know that it has been disproved since two counterexamples were found of the forms:

$$a^4 + b^4 + c^4 = d^4$$

and

$$a^5 + b^5 + c^5 + d^5 = e^5.$$

But before going any further, note that the expressions summoned in this paper are of geometrical shapes and not coordinates!

Try

And from those two counterexamples of the said forms comes my theory which states that the missing term to verify Euler's Conjecture is TIME.

Now that it's out of the way, my argument to support my theory is:

"Time is the only flexible dimensions, in other words: it can be equal to 0 while existing there and having some meaning, unlike spatial dimensions that when being equal to 0, lose their meaning."

This said, we got to clarify some additional points:

First:

It's true that we can say that the dimension of an expression is its number of terms, however, I have a new conjecture that I name **Amine's Recursive Triangle Representation (RTR)**, which states that we can represent any expression of the form:

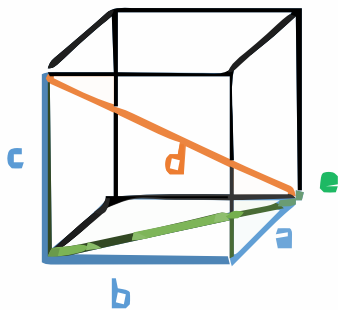
$$\sum_{i=0}^{n=2} x_i^n$$

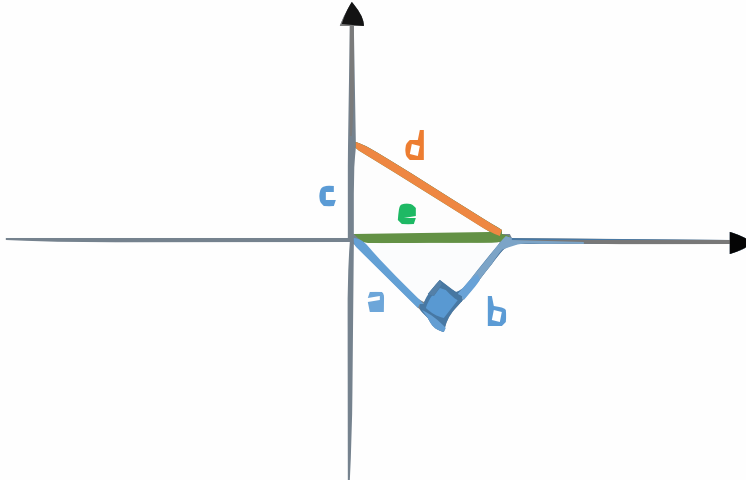
in a 2 axis graph all while keeping the angles and distances the same.

For example, we can represent the expression:

$$a^2 + b^2 + c^2 = d^2$$

which is the Pythagoras version in 3D, that is shown as in the following representation:





And we did that by substituting $a^2 + b^2$ as e^2 then replacing it in the expression:

$$e^2 + c^2 = d^2.$$

From which we conclude that powers also play a role in dimensions and not just terms do, and to clear the confusion, I don't really know why I put this thing up there, but still I think it is smoother to be put where it has been put.

This said, where is the time here? Well, well, well, it is either the additional absent term equal to 0 which means the expressions has 4 terms, OR, it is one of those terms, who knows? And there is nothing contradicting or illogical through the lenses of my theory in both cases.

Second:

If the number of terms is the absolute perfection, in other words it is equal to the power of the terms, then lads, same thing either time is already there watching in silence as a 0, or is there present as a value, all while nothing is illogical!

Third:

Now if the number of terms is less than the power by 1 term, my theory still holds, because it literally states that time is that missing term. However, if it is, someday found that an expression can have its number of terms less than its powers by 2 or more, then please, let this theory die peacefully in the humanity's wrong stuff archive except my dear RTR, take care of it ;).

Note

When I say time can have a real value, it can, because when analysing objects time does matter, like the position of something changes through time, and even in the quantum level – which I don't understand – time can be considered or be simplified into a spatial dimension. Yeah, I know I am blending math and physics and I may be just wrong as hell, but still why not?

Conclusion

This very modest theory tries to modestly relate Einstein's theory of relativity – or at least the part that states that time is the 4th dimension – to math, and takes it one step further by generalizing it to all dimensions – even theoretically –, all while introducing a novel concept into NT and math in general: TIME, reinforcing the idea that MATH IS REAL STUFF.

Please take care of my beloved RTR if this theory is doomed to be wrong.

Mohamed Amine Belachhab, 27/05/2025